

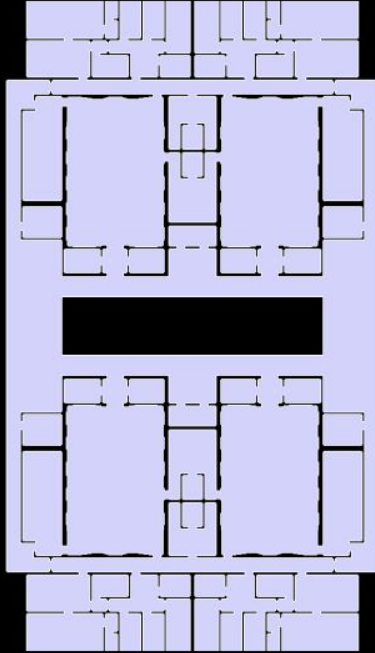
COURTHOUSE

GRAPH ANALYSIS

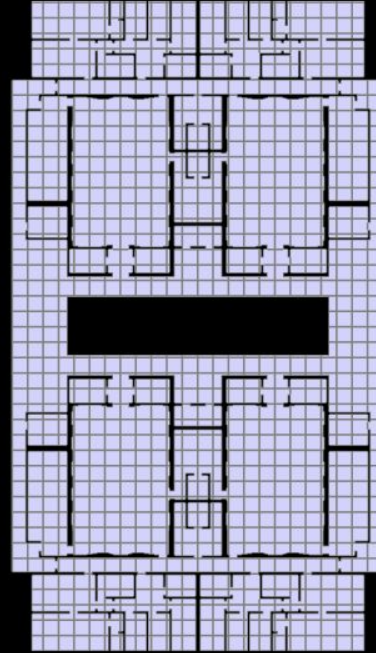
Graph Analysis of Architectural Structure

MaCAD · Graph Machine Learning Seminar · 2026

Building Overview . Courthouse Floor Plan



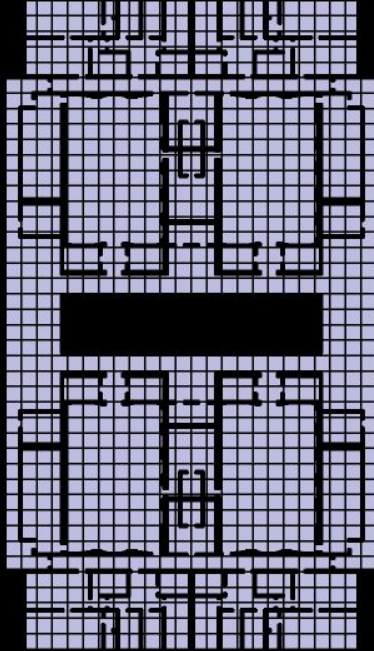
original floor plan geometry



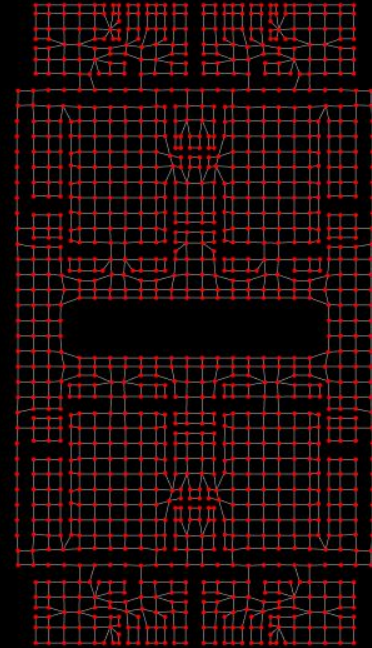
2m grid overlay applied

The courthouse plan is loaded as a BREP surface and treated as a single face. A uniform 2 m grid is then imposed across the bounding rectangle and clipped to the floor plan boundary, preparing the geometry for topological decomposition.

Methodology . Topological Shell & Analysis Graph



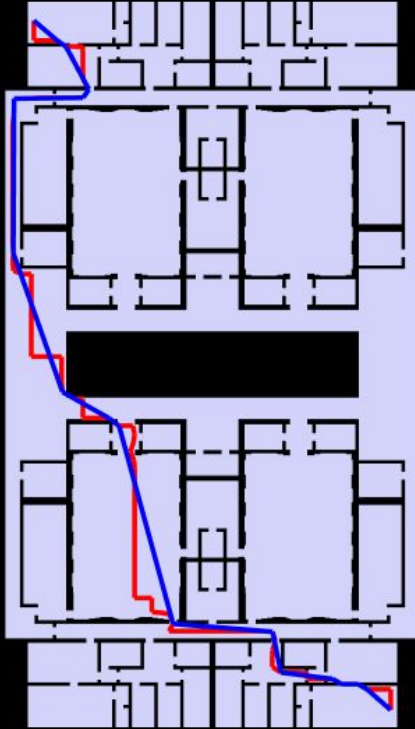
topological shell (sliced faces)



dual analysis graph

Shell: The floor plan is sliced with the grid to produce a shell of individual face cells. Each face inherits a unique `face_id` for metric tracking. Graph: A dual graph is derived – each face becomes a node, shared edges become connections.

01 . Shortest Path Analysis



red = topological path · blue = straightened path

1.23 seconds

Shortest Path Duration

134.38 m

Original Shortest Path Length

35.74 seconds

Straighten Wire Duration

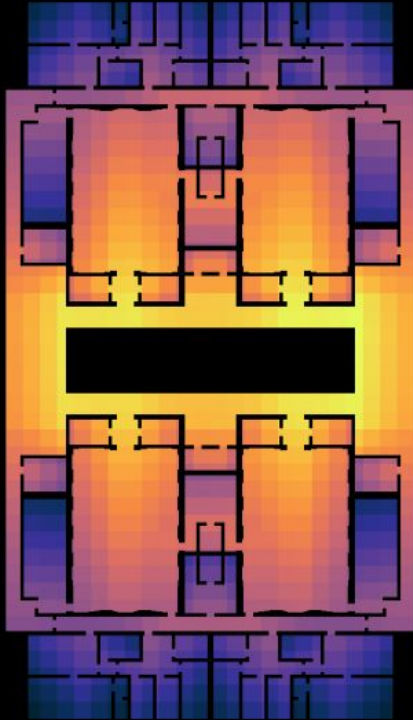
116.56 m

Straightened Shortest Path Length

Interpretation

The shortest walking path between extreme corners of the courthouse is 116.6 m, forced to detour around the central courtyard which acts as a circulation barrier separating north and south wings.

02 . Closeness Centrality (Global Integration)

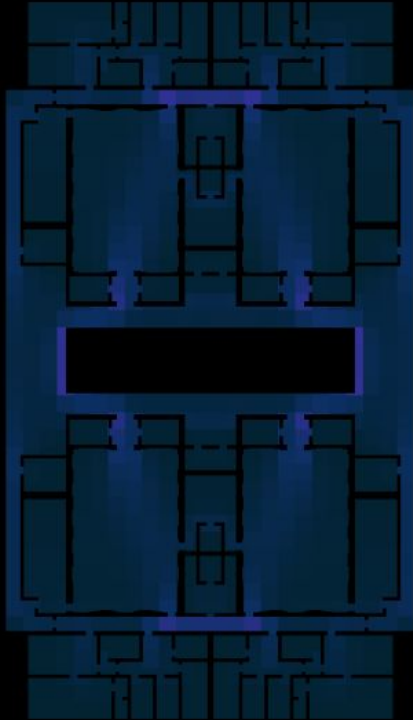


thermal scale: cool (low) → hot (high) integration

Interpretation

- High integration (warm zones) = spaces closest to all others - The ring around the courtyard scores high, marking it as the courthouse's circulation backbone.
- Low integration (cool zones) = peripheral or dead-end spaces - in a courthouse, candidates for chambers, holding, deliberation, or other privacy-sensitive functions.
- The metric corresponds to Space Syntax's global integration - empirically linked to pedestrian flow in urban networks, and indicative of natural circulation prominence in buildings.

03 . Betweenness Centrality (Choice / Bottleneck)

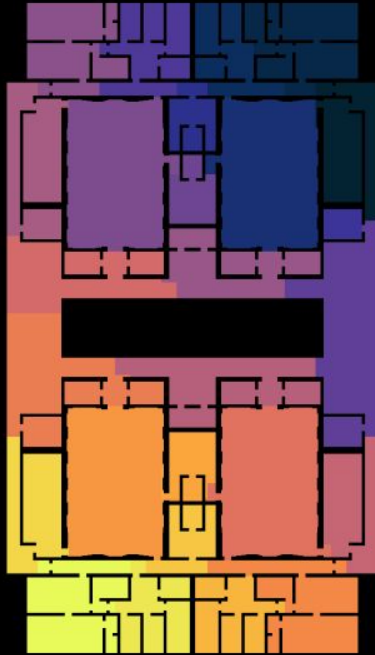


thermal scale: cool (low) → hot (high) betweenness

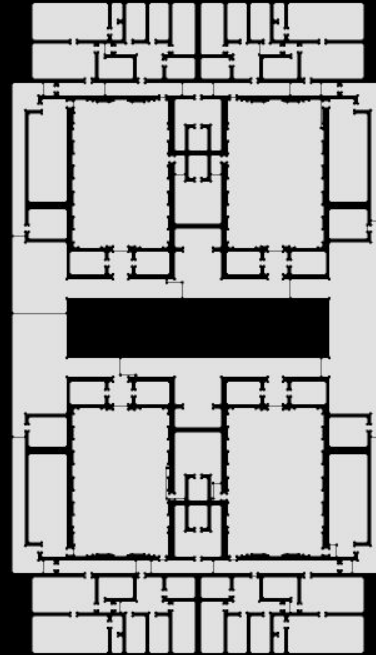
Interpretation

- High betweenness = critical connectors lying on the most shortest paths; removal would fragment circulation.
- These spaces act as chokepoints or strategic transitions - in a courthouse, the doorways and corridor links that funnel cross-wing movement.
- Low betweenness = spaces bypassed by through-traffic, suitable for private or secure functions where uninterrupted occupancy matters.
- Betweenness reveals single points of failure - high-value sites for access control, security screening, and egress planning.

04 . Community Detection (Spatial Zoning)



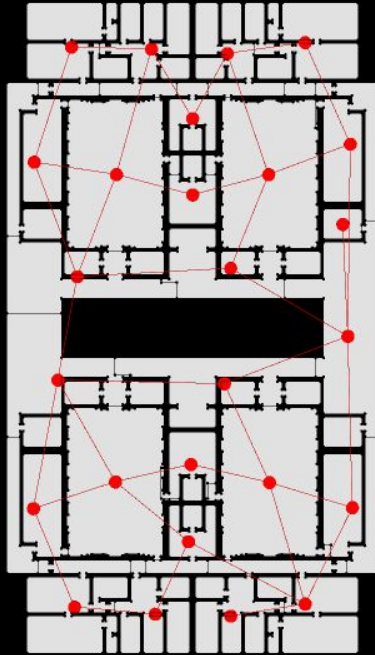
community partition (thermal color scale)



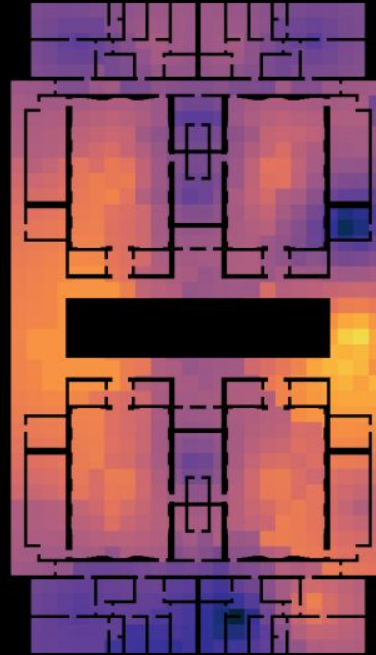
aggregated community shell

Louvain community detection groups cells by modularity maximisation, revealing natural spatial clusters without prior labelling. Each cluster is then merged into a single face, exposing a higher-level architectural zoning of the courthouse.

05 . Degree Centrality (Community Connectivity)



community-level graph overlay



degree centrality heat-map

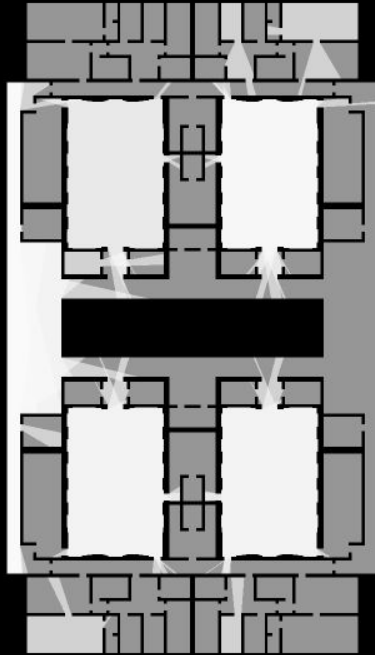
Degree centrality counts the number of neighbouring communities each zone shares an edge with. Hot (high-degree) zones are functional hubs that bridge multiple spatial clusters; cold zones are more self-contained or semi-private sectors of the building.

Key Findings

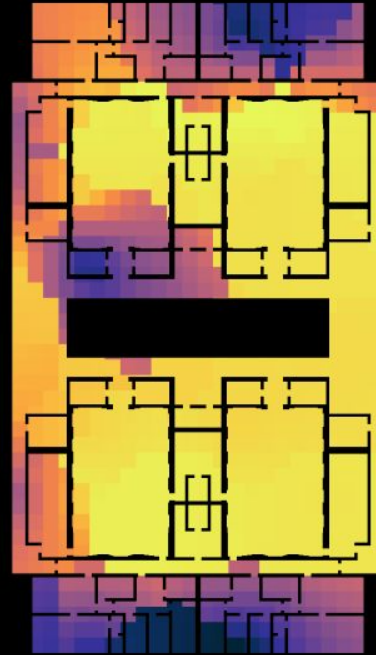
Key Findings

- Central corridors score highest in both closeness and betweenness, confirming their role as primary circulation spines.
- Community detection reveals 4–6 distinct spatial zones that map to functional areas (public, semi-public, restricted).
- High-degree community hubs correspond to transition spaces between public and judicial zones.
- Isovist analysis shows visibility concentrates around open atria and corridor intersections.
- Shortest paths confirm the upper-left entrance provides the most efficient diagonal routing.

Isovist & Visibility Graph Analysis Trial



isovist grid (10m spacing)



isovist face visibility (10m spacing)

At 10m spacing, the viewpoint grid is coarser than ideal; finer spacing would smooth transition zones and resolve sub-room visibility variations, but the dominant pattern (courtyard-organised, four-room dominance, peripheral drop off) is robust to grid resolution